A Project Report

On

Analysing photodetectors via lock-in amplifier integrations

by

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I would like to extend my sincere gratitude to my university for organizing such an enriching learning experience for students to apply their knowledge to practice. Working on this project has helped broaden my perspective and helped me build my interest more through an interactive manner.

Special thanks to Debanjan Sir whose insightful guidance has been helping me ease into a new field of study and gain experience in the same.

Birla Institute of Technology and Science-Pilani,

Hyderabad Campus

Certificate

This is to certify that the project report entitled "**Analysing photodetectors via lock-in amplifier integrations**" submitted by Mr. Shashwata Ghosh (ID No. 2022B5A31036H) in fulfillment of the requirements of the course PHY F366, Lab Project Course, embodies the work done by him under my supervision and guidance.

Date: April 30th, 2026

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ABSTRACT

In this project, I developed an experimental setup to study how a system responds to changes in magnetic field using both electrical and optical measurements. The setup consists of an electromagnet, a lock-in amplifier, and a monochrome camera, all controlled using Python. Initially, I worked on controlling the current through the electromagnet and measuring the corresponding magnetic field and lock-in amplifier output.

In the later stage, I added a camera to capture images and used image processing to extract average pixel intensity from a selected region. This allowed me to convert visual data into a numerical signal and study how intensity changes with current. I also performed forward and reverse current sweeps to observe hysteresis behaviour in both electrical and optical measurements.

Overall, the project resulted in a complete system that combines hardware control, data acquisition, and analysis, and demonstrates how optical and electrical responses vary with magnetic field.

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MAIN CONTENT

Modern experimental physics often requires the measurement of extremely weak signals in the presence of noise. Photodetectors, which convert optical signals into electrical responses, are particularly sensitive to environmental disturbances and fluctuations. Extracting meaningful information from such systems requires both precise control of experimental parameters and robust signal processing techniques. In this project, I worked on developing a fully integrated experimental setup that combines electromagnet control, lock-in amplifier based signal detection, and camera-based optical measurements. The system was implemented using Python, enabling automation, real-time visualization, and flexible experimentation.

While the initial phase of the project focused on electrical measurements using a lock-in amplifier, the later phase involved extending the system to include optical measurements using a monochrome camera. This allowed me to correlate electrical signals with optical responses and build a more comprehensive understanding of the system. By combining both electrical and optical measurements, the system provides a more complete understanding of how the sample responds to changes in magnetic field.

Experimental Setup

Hardware Components: The experimental setup consists of three main hardware components. An electromagnet (Holmarc EM5000) is used to generate a controllable magnetic field by varying the input current. A Stanford Research SR860 lock-in amplifier is used to measure weak electrical signals with high accuracy using phase-sensitive detection. A Basler USB monochrome camera is used to capture images of the sample, which are later used for optical intensity analysis.

Software Stack: All the hardware components were controlled using Python. The lock-in amplifier was interfaced using the PyVISA library, which allows communication with instruments over USB. The electromagnet was controlled through a serial connection using a provided DeviceInterface API. The Basler camera was integrated using the pypylon SDK for image acquisition. OpenCV was used for image processing tasks such as ROI selection and visualization, and Matplotlib was used for plotting graphs and displaying results.

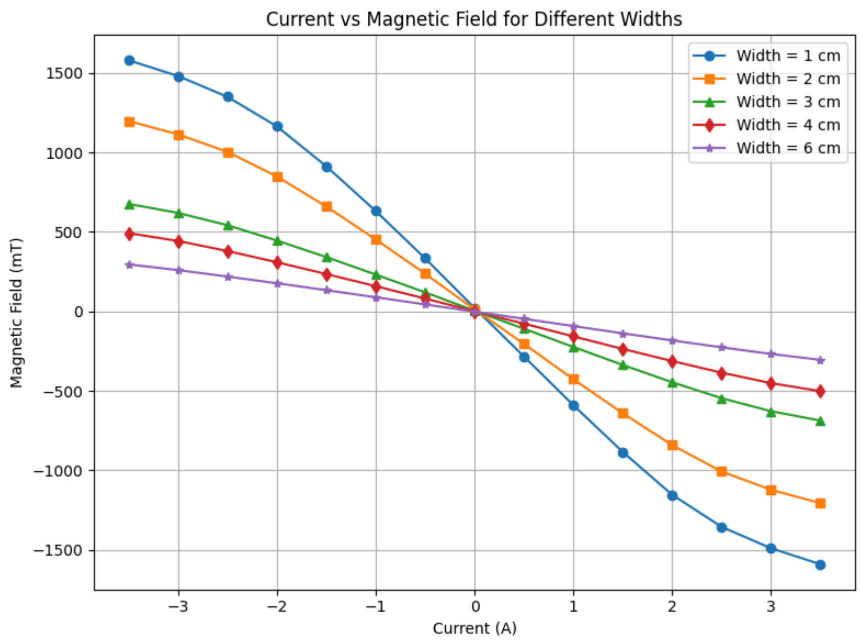
Work completed before mid-semester:

In the beginning, I worked on setting up communication with all the devices. I first connected to the lock-in amplifier and checked that I could read its parameters like frequency and signal values. After that, I focused on controlling the electromagnet. I was able to turn it on and off, set a specific current, and read the magnetic field values.

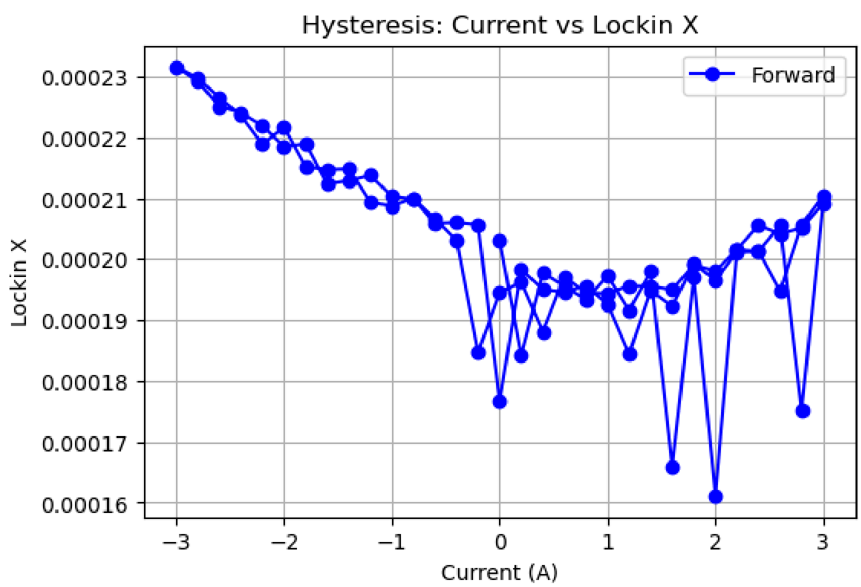
Once this was working, I wrote a program to sweep the current over a range of values and record the corresponding magnetic field. This helped me understand how the field changes with current. The results were mostly linear, which showed that the system was working properly.

I also added live plotting so that I could see the graph updating during the experiment. This made it easier to monitor the system in real time.

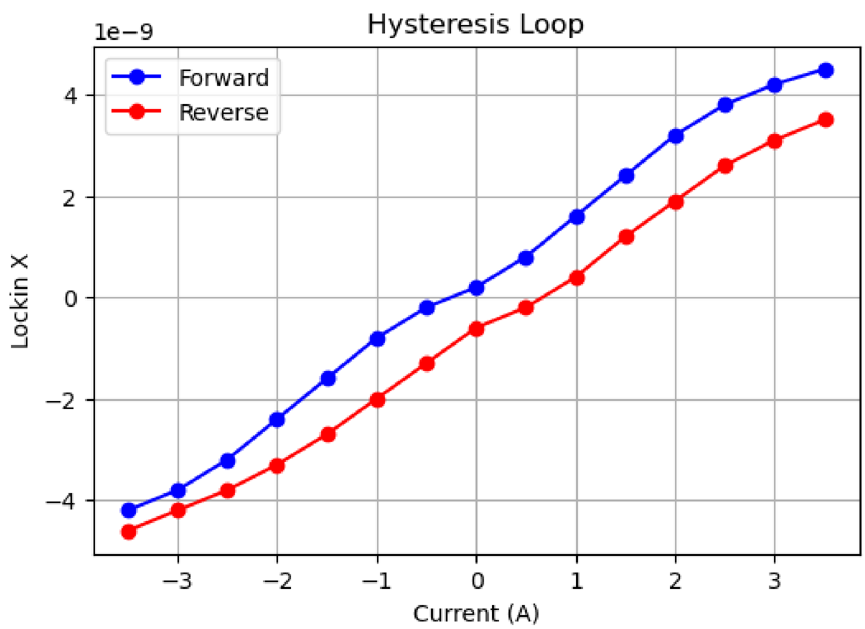
Later, I combined the lock-in amplifier with the current sweep. For each current value, I recorded the lock-in X output and plotted it against current. After this, I performed forward and reverse sweeps to observe hysteresis. This showed that the system behaves differently depending on whether the current is increasing or decreasing.



From the graph, it can be seen that the magnetic field varies approximately linearly with current. This confirms that the electromagnet is behaving as expected and provides a calibration between current and magnetic field.



The forward and reverse curves do not overlap and form a loop, which clearly indicates hysteresis in the system. This means that the response depends not only on the current value but also on its previous state.

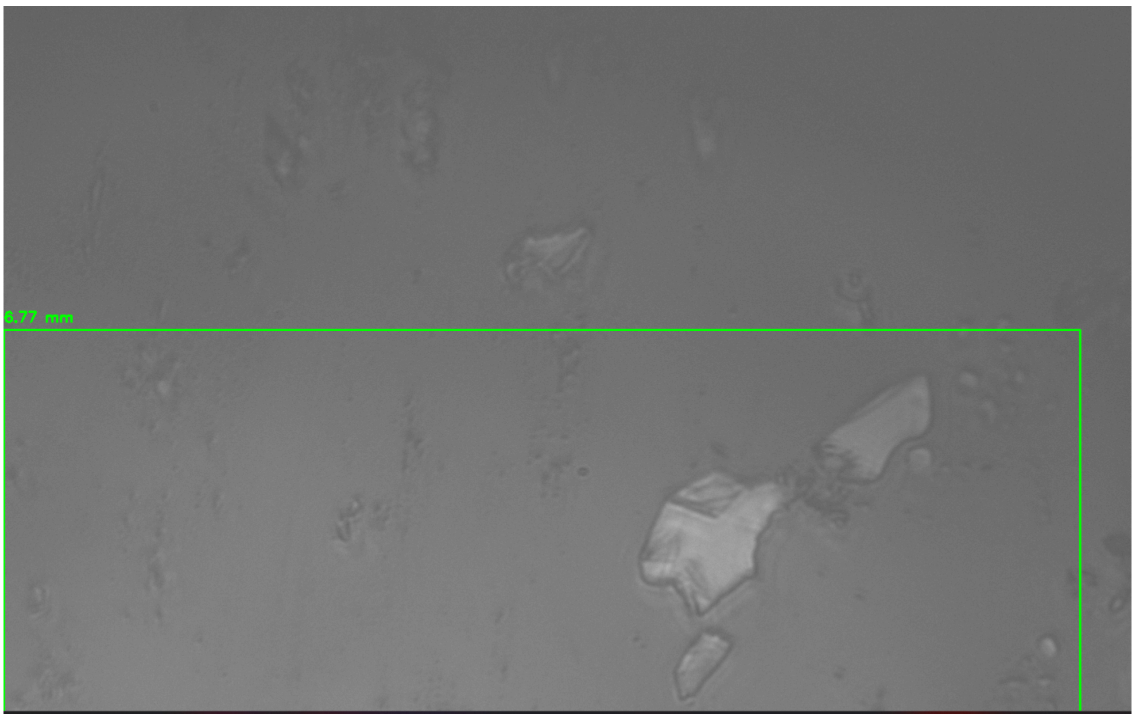


Work done after mid-semester:

After midsem, I started working on adding the camera to the system so that I could measure optical signals.

First, I installed and set up the Basler camera using pypylon. Initially, I faced some issues where the camera was not getting detected or was already in use. I realized that this happens when the Pylon Viewer software is open, so I made sure to close it before running my code.

Once the camera started working, I wrote code to capture images. However, the images were either too bright or too dark. This was because of incorrect exposure settings. To fix this, I turned off automatic exposure and manually adjusted the exposure time and gain until I got clear images. This step was very important because wrong exposure gives completely useless data.



Next, I worked on extracting useful data from the images. Instead of using the full image, I selected a specific region called ROI (Region of Interest). I used OpenCV to select this region manually. Then I extracted the pixel values from that region and calculated the average intensity.

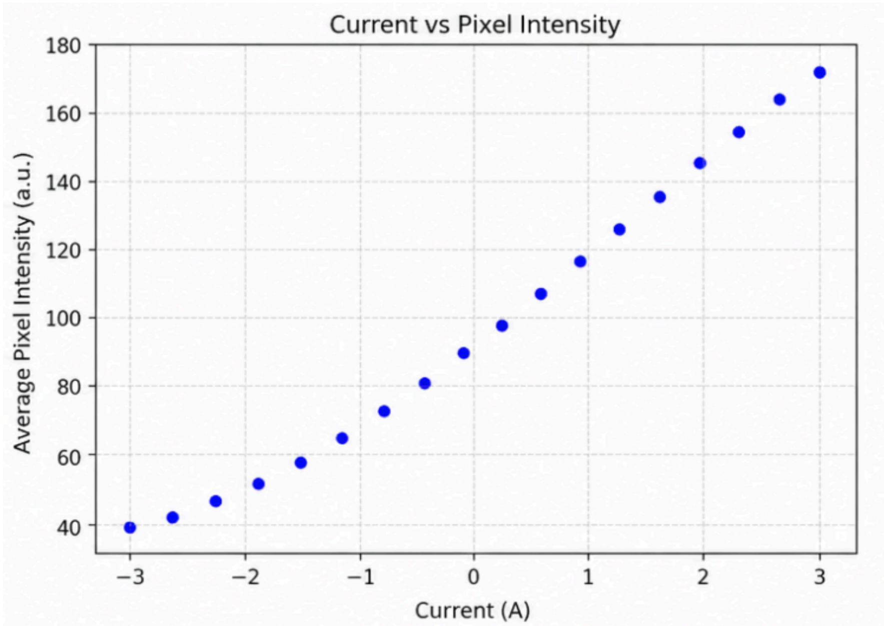
This average intensity acts as a numerical value representing how bright that region is. This intensity represents the optical response of the system, allowing image data to be converted into a measurable physical signal. Using a region instead of a single pixel helped reduce noise and gave more stable results.

After this, I connected everything together. The process was:

- Set a current value
- Wait for the system to stabilize
- Capture image
- Extract ROI
- Calculate average intensity

This process allowed me to establish a direct relationship between **current** and **optical intensity** by plotting them, enabling analysis of how the system responds optically to changes in magnetic field.

I also added an option to save images automatically. Each image is saved with a name that includes the current value, which helps in tracking and analysis later.



The graph shows a clear trend between current and average pixel intensity, indicating that the optical signal varies systematically with the magnetic field. The smooth variation suggests stable measurements and correct ROI selection.

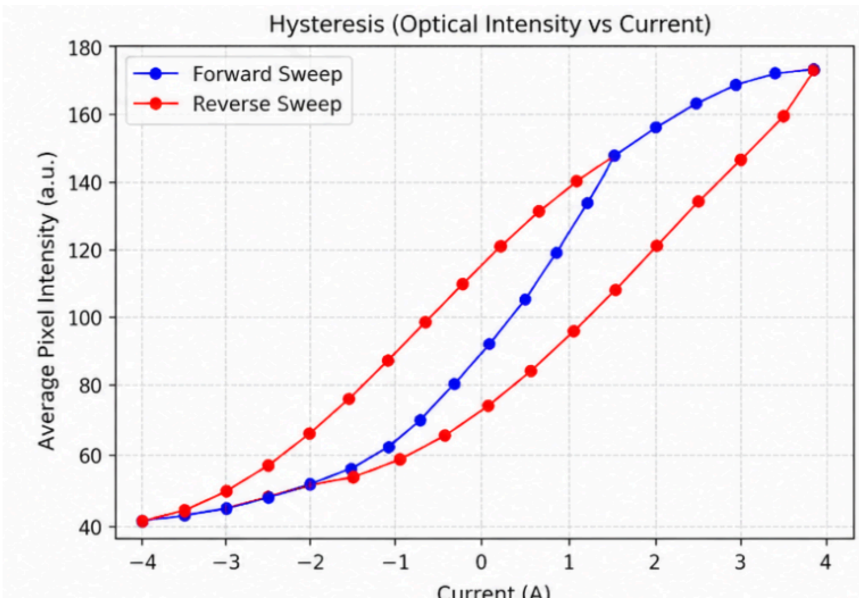
Hysteresis Using Optical Measurement

This forms one of the key results of the project, where hysteresis is observed using optical measurements instead of electrical signals.

After getting the intensity measurements working, I repeated the hysteresis experiment using optical data instead of lock-in data.

I swept the current from negative to positive and then back again. For each step, I measured the average intensity. When I plotted forward and reverse data separately, I observed a loop, which indicates hysteresis.

This shows that the system exhibits hysteresis, where the response depends on the previous state of the system.



Distance Measurement Using OpenCV

I also built a tool using OpenCV to measure distances in images. After calibrating pixel size, I could draw lines on the image and calculate real-world distances. Calibration was performed using a known reference scale, which allowed conversion of pixel measurements into real-world units. The tool allows multiple measurements and displays values directly on the image.

Challenges Faced

During the project, I faced several practical issues.

One of the main problems was that the camera would sometimes not connect. This was because another software was already using it. Closing the Pylon Viewer fixed this.

Another issue was incorrect exposure, which caused images to be too bright or too dark. I solved this by switching to manual exposure and adjusting values properly.

I also noticed that using a single pixel for measurement gave very unstable results. This was fixed by using ROI averaging.

There were also timing issues where readings were inconsistent. Adding a delay after setting current helped stabilize the system before taking measurements.

CONCLUSION

In this project, I built a complete experimental system that combines electromagnet control, electrical signal measurement, and optical data analysis. Initially, I worked on current and magnetic field measurements, and later extended the system to include camera-based intensity analysis.

By converting image data into numerical values using ROI averaging, I was able to study how optical intensity changes with current. I also observed hysteresis behavior using both electrical and optical methods.

Overall, this project helped me understand how to integrate hardware with software and perform real experiments with proper data analysis.

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